REPORT OF NOISE MONITORING
ENVIRONMENT MONITORING FOR KU CAMPUS
ABU DHABI, UAE

Request No.	**EQ21001733**
Report No.	**ER22000028**
Revision No.	0
Date	14/01/2022

PREPARED FOR:
Khalifa University for Science and Technology

Revision History					
0	14/01/2022	First Issue	QO	SR	RA
Revision No.	**Date**	**Description**	**Prepared**	**Checked**	**Approved**

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Request No: EQ21001733
Report No: ER22000028
Report Date: 14/01/2022

SUBJECT	:	NOISE MONITORING REPORT
PROJECT	:	Environment Monitoring for KU Campus
PROJECT NO	:	Not Applicable
OWNER	:	Khalifa University
CONSULTANT	:	Not Applicable
LOCATION	:	Khalifa University SAN Campus, Abu Dhabi
TESTING DATES	:	22-23/12/2021

1.0 INTRODUCTION

Pursuant to your request, ref. RFQ 6391231 dated 12th July 2021 related to the above subject, Arab Center for Engineering Studies is pleased to submit the obtained results of noise level monitoring carried out on the above project on 22-23/12/2021.

2.0 PURPOSE

The purpose of the study is to measure the Noise level during a specific monitoring period and assess its compliance with requirements of the Abu Dhabi Occupational Safety and Health System Framework (OSHAD-SF), Code of Practice, Cop3.0, Occupational Noise, Version 3.0, App.2

3.0 SCOPE OF WORKS

The scope of work consists of the following:

- Conducting noise level measurement by taking sound level meter readings on minutes basis for monitoring period and location(s) specified by Client (see Cl.4 and Cl.5 for the locations and monitoring details).
- Estimate of the "day time" averages Occupational Noise Exposure Limits and comparing with requirements of the Abu Dhabi Occupational Safety and Health System Framework (OSHAD-SF), Code of Practice, Cop3.0, Occupational Noise, Version 3.0, App.2.

4.0 LOCATION(S) AND SAMPLING DETAILS

Location Description	Visible Noise Sources Present	Sampling Height
Building 5 Library (Habshan)	-	1.5m above the ground level
Building 3 Laboratory (3027)	-	1.5m above the ground level
Building 5 Conference Room (229)	-	1.5m above the ground level
Building 3 (3013)	-	1.5m above the ground level
Building 5 (314)	-	1.5m above the ground level
Building 3 Laboratory (3011)	-	1.5m above the ground level
Building 4 (4012)	-	1.5m above the ground level
Building 3 Laboratory (3132)	-	1.5m above the ground level
Building 2 Laboratory (2023)	-	1.5m above the ground level
Building 2 (2016)	-	1.5m above the ground level
Building 1 Laboratory 112	-	1.5m above the ground level
Building 1 (013)	-	1.5m above the ground level
Arzanah Laboratory (8013)	-	1.5m above the ground level
Arzanah Laboratory (8130)	-	1.5m above the ground level
Arzanah Laboratory (8218)	-	1.5m above the ground level
ADNOC School	-	1.5m above the ground level
Umm Lulu Tower 1 Building	-	1.5m above the ground level
Building 7 (Spinneys)	-	1.5m above the ground level
Building 21 (101)	-	1.5m above the ground level
Mechanical Workshop M001	Exhaust fan, Air Conditioner	1.5m above the ground level

**Photo of monitoring location(s) are presented in Appendix A.*

5.0 MONITORING DETAILS

Location Description	Parameter	Sampling Dates and Time	Data Capture Rate%
Building 5 Library (Habshan)	**Background Noise Level L_{90A}**	22/12/2021 08:46 – 08:55	100%
	Source Noise L_{Aeq}	22/12/2021 08:56 – 09:10	100%
Building 3 Laboratory (3027)	**Background Noise Level L_{90A}**	22/12/2021 10:24 – 10:33	100%
	Source Noise L_{Aeq}	22/12/2021 10:34 – 10:48	100%
Building 5 Conference Room (229)	**Background Noise Level L_{90A}**	22/12/2021 09:26 – 09:36	100%
	Source Noise L_{Aeq}	22/12/2021 09:36 – 09:50	100%
Building 3 (3013)	**Background Noise Level L_{90A}**	22/12/2021 11:04 – 11:13	100%
	Source Noise L_{Aeq}	22/12/2021 11:14 – 11:28	100%
Building 5 (314)	**Background Noise Level L_{90A}**	22/12/2021 10:06 – 10:15	100%
	Source Noise L_{Aeq}	22/12/2021 10:16 – 10:30	100%
Building 3 Laboratory (3011)	**Background Noise Level L_{90A}**	22/12/2021 11:51 – 12:00	100%
	Source Noise L_{Aeq}	22/12/2021 12:01 – 12:15	100%
Building 4 (4012)	**Background Noise Level L_{90A}**	22/12/2021 10:51 – 11:00	100%
	Source Noise L_{Aeq}	22/12/2021 11:01 – 11:15	100%
Building 3 Laboratory (3132)	**Background Noise Level L_{90A}**	22/12/2021 12:22 – 12:31	100%
	Source Noise L_{Aeq}	22/12/2021 12:32 – 12:46	100%
Building 2 Laboratory (2023)	**Background Noise Level L_{90A}**	22/12/2021 12:45 – 12:55	100%
	Source Noise L_{Aeq}	22/12/2021 12:56 – 13:10	100%
Building 2 (2016)	**Background Noise Level L_{90A}**	22/12/2021 13:17 – 13:26	100%
	Source Noise L_{Aeq}	22/12/2021 13:27 – 13:41	100%
Building 1 Laboratory 112	**Background Noise Level L_{90A}**	22/12/2021 13:52 – 14:02	100%
	Source Noise L_{Aeq}	22/12/2021 14:02 – 14:17	100%
Building 1 (013)	**Background Noise Level L_{90A}**	22/12/2021 14:16 – 14:25	100%
	Source Noise L_{Aeq}	22/12/2021 14:26 – 14:40	100%

Arzanah Laboratory (8013)	**Background Noise Level L_{90A}**	22/12/2021 14:30 – 14:40	100%
	Source Noise L_{Aeq}	22/12/2021 14:41 – 14:55	100%
Arzanah Laboratory (8130)	**Background Noise Level L_{90A}**	22/12/2021 15:21 – 15:30	100%
	Source Noise L_{Aeq}	22/12/2021 15:31 – 15:45	100%
Arzanah Laboratory (8218)	**Background Noise Level L_{90A}**	22/12/2021 15:55 – 16:05	100%
	Source Noise L_{Aeq}	22/12/2021 16:06 – 16:20	100%
ADNOC School	**Background Noise Level L_{90A}**	23/12/2021 07:45 – 07:55	100%
	Source Noise L_{Aeq}	23/12/2021 07:56 – 08:10	100%
Umm Lulu Tower 1 Building	**Background Noise Level L_{90A}**	23/12/2021 08:40 – 08:50	100%
	Source Noise L_{Aeq}	23/12/2021 08:50 – 09:05	100%
Building 7 (Spinneys)	**Background Noise Level L_{90A}**	23/12/2021 09:51 – 10:00	100%
	Source Noise L_{Aeq}	23/12/2021 10:01 – 10:15	100%
Building 21 (101)	**Background Noise Level L_{90A}**	23/12/2021 11:23 – 11:32	100%
	Source Noise L_{Aeq}	23/12/2021 11:33 – 11:47	100%
Mechanical Workshop M001	**Background Noise Level L_{90A}**	23/12/2021 12:17 – 12:26	100%
	Source Noise L_{Aeq}	23/12/2021 12:27 -12:41	100%

6.0 MONITORING METHOD

Sound Level measurements were made with Sinus Tango Precision Sound Level Meter, Class 1 in accordance with ACES SOP047 based on ISO1996. The sound level meter was set at fast response in A-weighted frequency mode over the specified measurement period of 15 minutes. A one-half-inch microphone was mounted on a tripod approximately 1.5m above ground level with microphone pointed at the proper angle relative to an imaginary line to the source. Care was exercised to assure the measurement position was free from excess reflections from walls, columns, or other obstructing objects that may cause interference, and as much as possible, from heavy winds and other adverse conditions.

The analyzer and calibrator have valid annual calibration performed by ISO17025 accredited laboratory. On site calibrations were also performed before and after each measurement series. The calibration was done at 94dB and the calibration level variation was below ± 1.0 dB. (In the event that the deviation from the first calibration is greater than 1 dB (A) then the results obtained will not be valid and the assessment will have to be repeated).

Background noise assessment LA90 was made as much as possible in the absence of source noise. The assessment on the noise from the source under concern was completed by measurement of equivalent A-weighted sound level (LAeq) at each measurement location for a period of 15 minutes.

7.0 SUMMARY OF MONITORING RESULTS

NOISE RESULTS

Start / End Time: 22/12/2021 08:56 – 22/12/2021 09:10

Location	LA_{10}	LA_{90}	LAE	LAF_{max}	LAF_{min}	L(A)eq	LC_{peak}	LAS	LAS_{max}	LAS_{min}
	dB (A)									
Building 5 Library (Habshan)	-	34 (Background Noise)	-	-	-	38	-	-	-	-
*Noise Limits db(A), See [Ref.]**	-	-	-	-	-	*100 Max. allowable*		-	-	-

Start / End Time: 22/12/2021 10:34 – 22/12/2021 10:48

Location	LA_{10}	LA_{90}	LAE	LAF_{max}	LAF_{min}	L(A)eq	LC_{peak}	LAS	LAS_{max}	LAS_{min}
	dB (A)									
Building 3 Laboratory (3027)	-	55 (Background Noise)	-	-	-	59	-	-	-	-
*Noise Limits db(A), See [Ref.]**	-	-	-	-	-	*100 Max. allowable*		-	-	-

Start / End Time: 22/12/2021 09:36 – 22/12/2021 09:50

Location	LA_{10}	LA_{90}	LAE	LAF_{max}	LAF_{min}	L(A)eq	LC_{peak}	LAS	LAS_{max}	LAS_{min}
	dB (A)									
Building 5 Conference Room (229)	-	37 (Background Noise)	-	-	-	42	-	-	-	-
*Noise Limits db(A), See [Ref.]**	-	-	-	-	-	*100 Max. allowable*		-	-	-

Start / End Time: 22/12/2021 11:14 – 22/12/2021 11:28

Location	LA_{10}	LA_{90}	LAE	LAF_{max}	LAF_{min}	L(A)eq	LC_{peak}	LAS	LAS_{max}	LAS_{min}
	dB (A)									
Building 3 (3013)	-	34 (Background Noise)	-	-	-	34	-	-	-	-
*Noise Limits db(A), See [Ref.]**	-	-	-	-	-	*100 Max. allowable*		-	-	-

Start / End Time: 22/12/2021 10:16 – 22/12/2021 10:30

Location	LA_{10}	LA_{90}	LAE	LAF_{max}	LAF_{min}	L(A)eq	LC_{peak}	LAS	LAS_{max}	LAS_{min}
	dB (A)									
Building 5 (314)	-	37 (Background Noise)	-	-	-	50	-	-	-	-
*Noise Limits db(A), See [Ref.]**	-	-	-	-	-	*100 Max. allowable*		-	-	-

Start / End Time: 22/12/2021 12:01 – 22/12/2021 12:15

Location	LA_{10}	LA_{90}	LAE	LAF_{max}	LAF_{min}	L(A)eq	LC_{peak}	LAS	LAS_{max}	LAS_{min}
	dB (A)									
Building 3 Laboratory (3011)	-	57 (Background Noise)	-	-	-	58	-	-	-	-
*Noise Limits db(A), See [Ref.]**	-	-	-	-	-	*100 Max. allowable*		-	-	-

Start / End Time: 22/12/2021 11:01 – 22/12/2021 11:15

Location	LA_{10}	LA_{90}	LAE	LAF_{max}	LAF_{min}	L(A)eq	LC_{peak}	LAS	LAS_{max}	LAS_{min}
	dB (A)									
Building 4 (4012)	-	28 (Background Noise)	-	-	-	33	-	-	-	-
*Noise Limits db(A), See [Ref.]**	-	-	-	-	-	*100 Max. allowable*		-	-	-

Start / End Time: 22/12/2021 12:32 – 22/12/2021 12:46

Location	LA_{10}	LA_{90}	LAE	LAF_{max}	LAF_{min}	L(A)eq	LC_{peak}	LAS	LAS_{max}	LAS_{min}
	dB (A)									
Building 3 Laboratory (3132)	-	52 (Background Noise)	-	-	-	53	-	-	-	-
*Noise Limits db(A), See [Ref.]**	-	-	-	-	-	*100 Max. allowable*		-	-	-

Start / End Time: 22/12/2021 12:56 – 22/12/2021 13:10

Location	LA_{10}	LA_{90}	LAE	LAF_{max}	LAF_{min}	L(A)eq	LC_{peak}	LAS	LAS_{max}	LAS_{min}
	dB (A)									
Building 2 Laboratory (2023)	-	48 (Background Noise)	-	-	-	49	-	-	-	-
*Noise Limits db(A), See [Ref.]**	-	-	-	-	-	*100 Max. allowable*		-	-	-

Start / End Time: 22/12/2021 13:27 – 22/12/2021 13:41

Location	LA_{10}	LA_{90}	LAE	LAF_{max}	LAF_{min}	L(A)eq	LC_{peak}	LAS	LAS_{max}	LAS_{min}
	dB (A)									
Building 2 (2016)	-	35 (Background Noise)	-	-	-	37	-	-	-	-
*Noise Limits db(A), See [Ref.]**	-	-	-	-	-	*100 Max. allowable*		-	-	-

Start / End Time: 22/12/2021 14:02 – 22/12/2021 14:17

Location	LA_{10}	LA_{90}	LAE	LAF_{max}	LAF_{min}	L(A)eq	LC_{peak}	LAS	LAS_{max}	LAS_{min}
	dB (A)									
Building 1 Laboratory 112	-	33 (Background Noise)	-	-	-	37	-	-	-	-
*Noise Limits db(A), See [Ref.]**	-	-	-	-	-	*100 Max. allowable*		-	-	-

Start / End Time: 22/12/2021 14:26 – 22/12/2021 14:40

Location	LA_{10}	LA_{90}	LAE	LAF_{max}	LAF_{min}	L(A)eq	LC_{peak}	LAS	LAS_{max}	LAS_{min}
	dB (A)									
Building 1 (013)	-	33 (Background Noise)	-	-	-	35	-	-	-	-
*Noise Limits db(A), See [Ref.]**	-	-	-	-	-	*100 Max. allowable*		-	-	-

Start / End Time: 22/12/2021 14:41 – 22/12/2021 14:55

Location	LA_{10}	LA_{90}	LAE	LAF_{max}	LAF_{min}	L(A)eq	LC_{peak}	LAS	LAS_{max}	LAS_{min}
	dB (A)									
Arzanah Laboratory (8013)	-	31 (Background Noise)	-	-	-	49	-	-	-	-
*Noise Limits db(A), See [Ref.]**	-	-	-	-	-	*100 Max. allowable*		-	-	-

Start / End Time: 22/12/2021 15:31 – 22/12/2021 15:45

Location	LA_{10}	LA_{90}	LAE	LAF_{max}	LAF_{min}	L(A)eq	LC_{peak}	LAS	LAS_{max}	LAS_{min}
	dB (A)									
Arzanah Laboratory (8130)	-	62 (Background Noise)	-	-	-	63	-	-	-	-
*Noise Limits db(A), See [Ref.]**	-	-	-	-	-	*100 Max. allowable*		-	-	-

Start / End Time: 22/12/2021 16:06 – 22/12/2021 16:20

Location	LA_{10}	LA_{90}	LAE	LAF_{max}	LAF_{min}	L(A)eq	LC_{peak}	LAS	LAS_{max}	LAS_{min}
	dB (A)									
Arzanah Laboratory (8218)	-	38 (Background Noise)	-	-	-	39	-	-	-	-
*Noise Limits db(A), See [Ref.]**	-	-	-	-	-	*100 Max. allowable*		-	-	-

Start / End Time: 23/12/2021 07:56 – 23/12/2021 08:10

Location	LA_{10}	LA_{90}	LAE	LAF_{max}	LAF_{min}	L(A)eq	LC_{peak}	LAS	LAS_{max}	LAS_{min}
	dB (A)									
ADNOC School	-	42 (Background Noise)	-	-	-	47	-	-	-	-
*Noise Limits db(A), See [Ref.]**	-	-	-	-	-	*100 Max. allowable*		-	-	-

Start / End Time: 23/12/2021 08:50 – 23/12/201 09:05

Location	LA_{10}	LA_{90}	LAE	LAF_{max}	LAF_{min}	L(A)eq	LC_{peak}	LAS	LAS_{max}	LAS_{min}
	dB (A)									
Umm Lulu Tower 1 Building	-	44 (Background Noise)	-	-	-	57	-	-	-	-
*Noise Limits db(A), See [Ref.]**	-	-	-	-	-	*100 Max. allowable*		-	-	-

Start / End Time: 23/12/2021 10:01 – 23/12/2021 10:15

Location	LA_{10}	LA_{90}	LAE	LAF_{max}	LAF_{min}	L(A)eq	LC_{peak}	LAS	LAS_{max}	LAS_{min}
	dB (A)									
Building 7 (Spinneys)	-	48 (Background Noise)	-	-	-	54	-	-	-	-
*Noise Limits db(A), See [Ref.]**	-	-	-	-	-	*100 Max. allowable*		-	-	-

Start / End Time: 23/12/2021 11:33 – 23/12/2021 11:47

Location	LA_{10}	LA_{90}	LAE	LAF_{max}	LAF_{min}	L(A)eq	LC_{peak}	LAS	LAS_{max}	LAS_{min}
	dB (A)									
Building 21 (101)	-	43 (Background Noise)	-	-	-	44	-	-	-	-
*Noise Limits db(A), See [Ref.]**	-	-	-	-	-	*100 Max. allowable*		-	-	-

Start / End Time: 23/12/2021 12:27 – 23/12/2021 12:41

Location	LA_{10}	LA_{90}	LAE	LAF_{max}	LAF_{min}	L(A)eq	LC_{peak}	LAS	LAS_{max}	LAS_{min}
	dB (A)									
Mechanical Workshop M001	-	62 (Background Noise)	-	-	-	63	-	-	-	-
*Noise Limits db(A), See [Ref.]**	-	-	-	-	-	*100 Max. allowable*		-	-	-

**[Ref]: Abu Dhabi Occupational Safety and Health System Framework (OSHAD-SF), Code of Practice, Cop3.0, Occupational Noise, Version 3.0, App.2*

METEOROLOGICAL DATA

Location Id No.	Ambient Temperature	Relative Humidity
	(Min./Max.)ºC	%
Building 5 Library (Habshan)	19.5 – 22.6	40
Building 3 Laboratory (3027)	19.2 – 22.1	42
Building 5 Conference Room (229)	22.5 – 24.1	39
Building 3 (3013)	20.4 – 24.5	44
Building 5 (314)	20.9 – 23.1	40
Building 3 Laboratory (3011)	20.2 – 23.9	43
Building 4 (4012)	21.1 – 24.2	41
Building 3 Laboratory (3132)	21.5 – 23.8	42
Building 2 Laboratory (2023)	21.2 – 24.2	41
Building 2 (2016)	20.6 – 24.3	45
Building 1 Laboratory 112	20.0 – 23..3	42
Building 1 (013)	21.3 – 23.7	44
Arzanah Laboratory (8013)	22.3 – 24.8	45
Arzanah Laboratory (8130)	20.5 – 24.3	45
Arzanah Laboratory (8218)	20.4 – 24.6	40
ADNOC School	20.0 – 23.9	40
Umm Lulu Tower 1 Building	21.2 – 24.7	41

Building 7 (Spinneys)	19.5 – 22.7	39
Building 21 (101)	21.1 – 24.0	40
Mechanical Workshop M001	20.7 – 33.9	41

*Ref: www.ncms.ae

8.0 IMPORTANT NOTES

- The results presented in this report relate only to the specific location and specific monitoring period. It should be noted that they may be subject to variation caused by variation of activities at/or nearby the site and/or change of environmental conditions.
- This report shall not be reproduced except in full, without written approval of ACES.
- Unless otherwise stated, Client and project details as well as sampling and sample identification information included in this report are provided by the Client. ACES shall not be liable for any incorrect results arising from unclear, erroneous, incomplete, misleading or false information provided by the Client.
- Specification Limits included in this report are provided for information, with no obligation to ACES for providing a statement of conformity.

9.0 APPENDICES

Appendix A - Photograph of Monitoring Location(s)

Appendix B - Project Resources

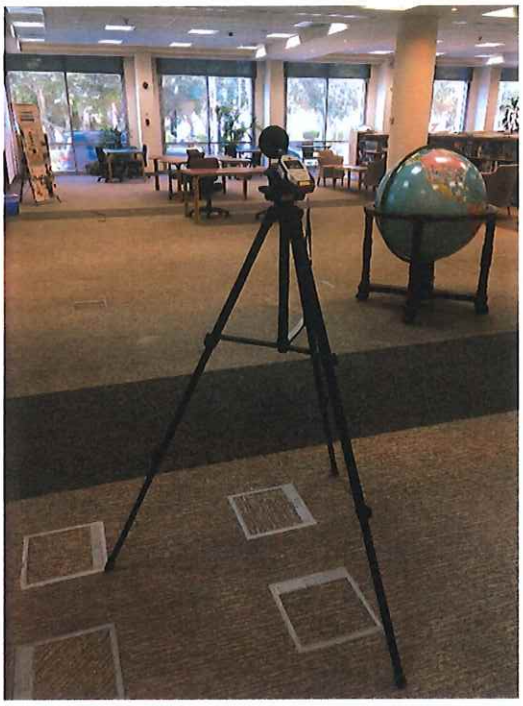
Appendix C - Definitions and Abbreviations Used

Sincerely yours,
ARAB CENTER FOR ENGINEERING STUDIES

Environmental and Chemical Department Manager

Arab Center For Engineering Studies
المركز العربي للدراسات الهندسية

APPENDIX A –PHOTOGRAPH OF MONITORING LOCATION(S)



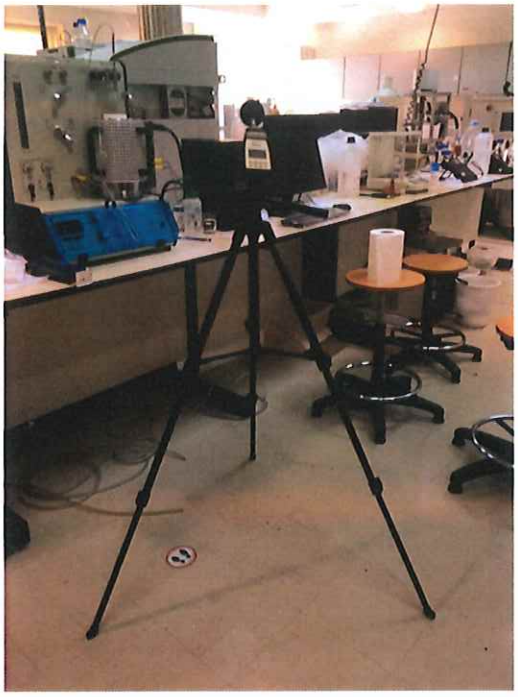
Building 5 Library (Habshan)

Building 3 Laboratory (3027)

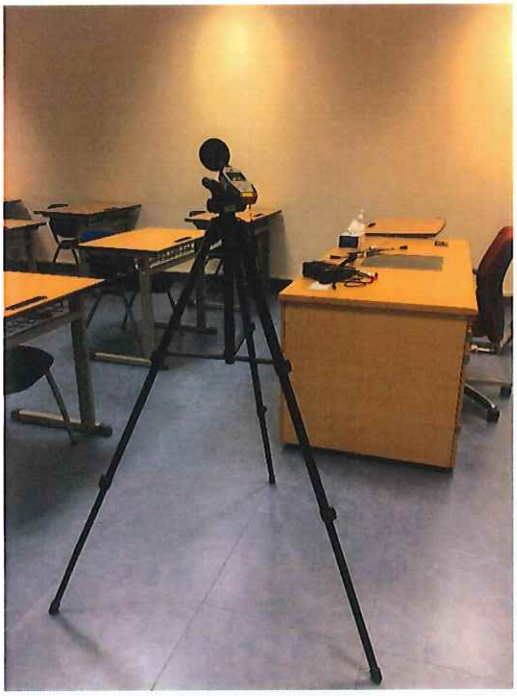
Building 5 Conference Room (229)

Building 3 (3013)

Building 5 (314)



Building 3 Laboratory (3011)

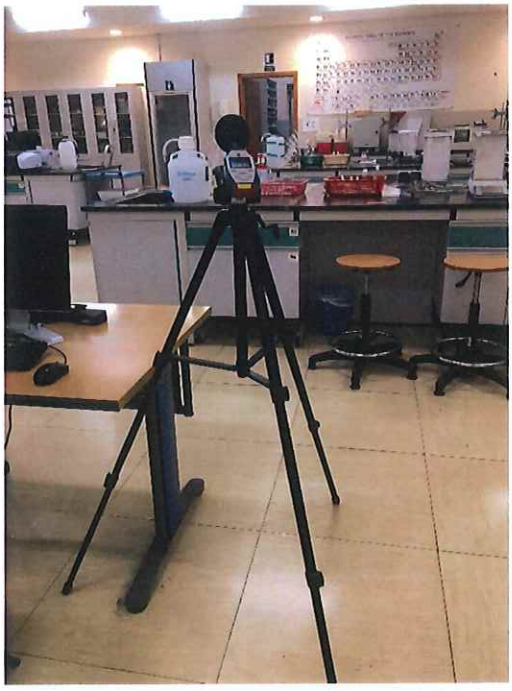


Building 4 (4012)

Building 3 Laboratory (3132)

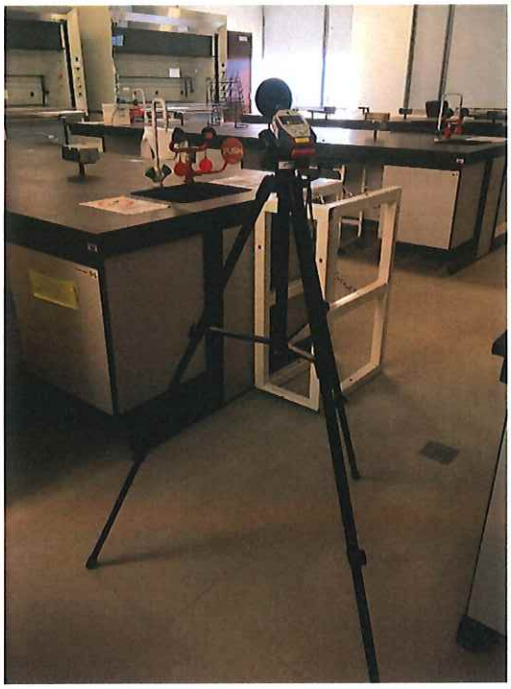
Building 2 Laboratory (2023)

Building 2 (2016)



Building 1 Laboratory 112

Building 1 (013)



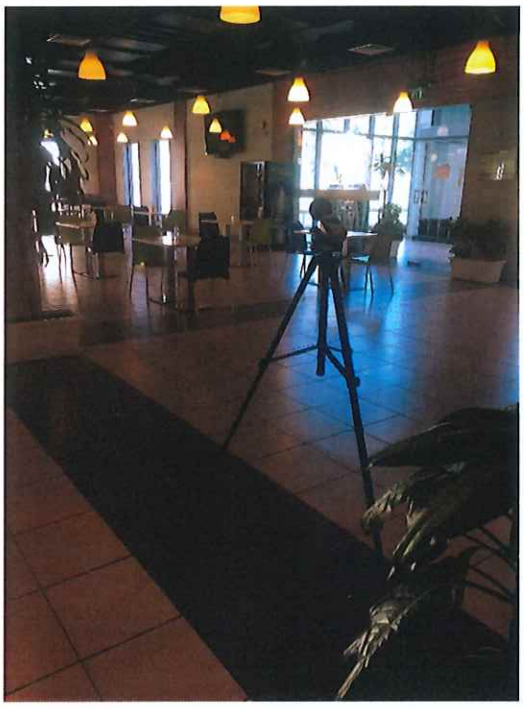
Arzanah Laboratory (8013)

Arzanah Laboratory (8130)

Arzanah Laboratory (8218)

ADNOC School

Umm Lulu Tower 1 Building



Building 7 (Spinneys)

Building 21 (101)

Mechanical Workshop M001

APPENDIX B

PROJECT RESOURCES

Equipment Used

Equipment	Equipment ID
Sound Level Meter	V24/1/4H-A
Calibrator	V24/1/4H-B
RH meter	V16/4/100H
Thermometer	V16/4/100H
Anemometer	V46/1/1H

Calibration Certificates of the above equipments are available up on request.

ACES Monitoring Personnel

Position	Name
Supervisor	Samir Elrifai
Environmental Monitoring Technician	Qudus Olatunde

APPENDIX C

DEFINITIONS AND ABBREVIATIONS USED

A Weighing	A standard weighting of the audible frequencies designed to reflect the response of the human ear
db (A)	Decibels A weighted
LAeq	Equivalent continuous sound pressure level. A measure of the average sound pressure level
LAF_{max}	The minimum sound level with A weighting
LAF_{min}	The maximum sound level with A weighting
LAF	Sound level with 'A' frequency weighting and fast time weighting
LAE	Sound Exposure Level (SEL) with 'A' frequency weighting
LC_{peak}	Peak sound pressure level with 'C' frequency weighting
LAS	Sound level with 'A' frequency weighting and slow time weighting
LAS_{max}	The maximum Sound Level with 'A', frequency weighting and Slow Time weighting
LAS_{min}	The minimum Sound Level with 'A', frequency weighting and Slow Time weighting
LA_{10} & LA_{90}	Background sound pressure level

-END OF REPORT-